

Current focus areas of T&M companies for academia

Apart from recommending industry-grade tools for colleges and universities, T&M vendors are introducing teaching kits and labs to ease the learning and teaching processes for students and academicians, respectively. Today, oscilloscopes are also embedded with training-mode features, enabling users to investigate and educate themselves.

From a product perspective, Keysight focuses on three areas for educational institutes. Pathy says, "One is basic equipment for faculty and students who can use and put together some experiments. Next are teaching labs that are a bundled solution, where we not only provide the equipment but also teaching kits. Last offering focuses on postgraduate (PG) curriculum and research in a chosen domain." He goes on to add, "For example, at Keysight we have equipment and design tools for anything to do with electronics, be it radio frequency (RF), wireless, nanotechnology, high-speed digital or semiconductor. If a college has a PG program on wireless communications, we have solutions where we supply the complete lab—equipment, teaching kits, design solutions and even structured training programs for faculty."

Companies like Tektronix and Rohde & Schwarz are focusing on incorporating an integrated courseware system and training mode into their instruments, respectively. To help students and professors acquire in-depth knowledge through their instruments, Scientech Technologies is working on high-end VLSI products and automated test equipment (ATE), which can help students get the actual feel of ATE testing processes.

Amol Kadu, regional manager-west region, Scientech Technologies Pvt Ltd, says, "Our main principle has always been to make students feel that engineering is not rocket science, by providing them with equipment that make their life easier while studying. These equipment are aimed to help teachers and students grasp the

subject effectively." He adds, "We are also upgrading our oscilloscopes that can be used as ATE rather than just conventional oscilloscopes. Our DesignLab is the first breakthrough in this segment of ATE."

Anritsu has partnered with Elliptika to come up with a product called Educitika, which is a microwave teaching kit for students at colleges and universities. Anritsu's VNA Master series of handheld vector network analysers is part of this solution, which can be utilised by academicians to enlighten students on fundamentals of RF and microwave technologies. This mostly serves the market outside India. For Indian market, Anritsu has developed an RF and microwave trainer kit in association with Foundation for Innovation and Technology, IIT Delhi.

New entrants in T&M equipment for education

There were quite a few industry-specific products released in the last one year, also suitable for academia and specifically just for educational institutions and universities. Select products have been talked about in detail below.

All-in-one instruments. There is a whole set of people designing circuits and systems, and making measurements with the help of benchtop instruments. Six months back, with the intent of making benchtop instrumentation easy, NI came up with a product called VirtualBench (refer to table for more information). "Even though technology has changed in different areas of automated test, advancements for benchtop measurements have been very limited," feels Mohanram. VirtualBench is a combination of a function generator, oscilloscope, digital multimeter, high-speed logic analyser and power supply integrated into a single equipment, programmable using LabVIEW. There is also an option for users to not program the device by simply using it as a plug-and-play instrument via USB or Wi-Fi connectivity.

With a ready-to-run application for Windows based PCs and iPads, users can control and view all measurements on the device's screen. Mohanram informs, "For the industry, it enables a generation of signals and measurement on a single device rather than having multiple instruments for multiple tasks. For academia, VirtualBench is available at a relatively lower price, which has helped a lot of electronics labs re-look at their current set-up and replace their existing box-type instruments." He adds, "This is one instrument that is helping academicians change the way they have been doing measurements in electronics, or any lab for that matter."

Scientech's PC based instrument, DesignLab, looks to make it easier for students with a seven-in-one equipment, which has a digital storage oscilloscope, function generator, 16-channel logic analyser, pattern and word generator, and variable power supply. Kadu says, "DesignLab helps students and professionals to have hands-on experience, realise proof of concepts and their projects." He adds, "Educational institutions, like IITs, NITs and some other engineering and polytechnic colleges, are users of Scientech products, as we have solidified their trust in our products over the years now."

Although expensive, Tektronix's MDO3000 series of mixed domain oscilloscopes benefit engineers and educational institutes into R&D by offering the functionality of six independent instruments, namely, spectrum analyser, logic analyser, protocol analyser, arbitrary function generator and digital voltmeter (DVM). On the flip-side, only three functions (oscilloscope, spectrum analyser and DVM) are standard, others cost extra.

Benchtop digital oscilloscopes. Keysight has some new introductions where they feel that the improved user interface will allow students to use the instrument in a much more effective manner. Pathy says, "We have introduced a scope called 3000T with a much better user interface." These 3000 X-Series oscilloscopes